

KENAR NARAYAKA

+44 7918 448877 kenar.narayaka@cs.ox.ac.uk linkedin.com/in/narayaka github.com/kxnar

Education

University of Oxford

MCompSciPhil Computer Science & Philosophy

Oct. 2025 – Jun. 2029

Christ Church

Loughborough College

*A-Level Mathematics, Further Mathematics, Computer Science (A*AA)*

Sep. 2023 – Jun. 2025

Loughborough, UK

Woodbrook Vale Secondary School

10 GCSEs (9 Grade 9s, 1 Grade 8); FSMQ Additional Mathematics (A)

Sep. 2018 – Jun. 2023

Loughborough, UK

Relevant Coursework

- Design & Analysis of Algorithms
- Discrete Mathematics
- Intro. Proof Systems
- Functional Programming
- Imperative Programming
- Probability
- Logic

Experience

Maths & Computer Science Tutor

Independent

April 2025 – Present

Remote / Loughborough, UK

- Delivered sustained 1-to-1 tutoring in mathematics and computer science for competitive university entrance exams, supporting a successful admission to a selective private university
- Designed targeted preparation integrating calculus, algebra, probability, and Python-based problem solving, with custom problem sets and timed practice

Hitachi Astemo

Backend Engineer Intern

June 2024 – August 2024

Jakarta, Indonesia

- Developed internal workflow systems in PHP that fully replaced paper-based HR processes, improving traceability and eliminating manual document handling across the department
- Designed and implemented an end-to-end digital request pipeline using PHP and the Fat-Free Framework, covering submission, validation, approval, and fulfilment
- Built automatic approval-routing logic, server-side form validation, PDF document generation, and direct network printing as core pipeline components
- Developed REST APIs and administrative tooling for managing high-volume employee records under strict data-consistency requirements
- Implemented role-based access control, restricting sensitive records and administrative operations to authorised staff
- Optimised SQL queries and ORM mappings to reduce latency and support faster updates across large employee datasets

Projects

3D Fractal Simulator | *JavaScript, three.js, GLSL* | *Loughborough College*

Feb. 2025

- Developed a real-time 3D fractal visualiser for A-Level Computer Science Non-Examined Assessment using JavaScript, three.js, and GLSL fragment shaders with raymarching and cone-marching (vertex shader) techniques.
- Achieved 69/70 score on OCR CS A-Level NEA, highest in school history.
- Implemented configurable fractals including Mandelbulb, Mandelcube, and Juliabulb with adjustable parameters for scale, iteration depth, morph speed, and camera behaviour
- Optimised GPU performance through shader-level mathematical simplifications, reduced computational overhead, and dynamic fractal iteration scaling for smoother real-time rendering when factoring camera movement.
- Designed interactive UI for live parameter manipulation, making complex mathematical structures more accessible and easier to explore in real time.

Activities

CapitOx Quant Academy

University of Oxford

Jan. 2026 – Present

- Selected for a competitive programme run with the Oxford Man Institute focused on systematic investing
- Completed training in portfolio optimisation, factor models, and multi-signal strategies
- Developed and backtested example investment strategy using Python and C++